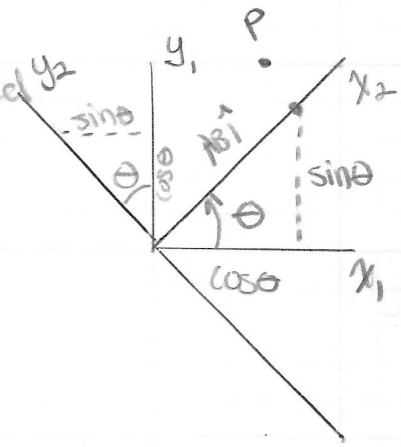
Coordinate Transforms

I Notation

- Same point to be represented in different ways based on the frame of reference.
- Pre-subscript to denote frame: ${}_AP$ (Point P in frame A)

II 2D Rotation

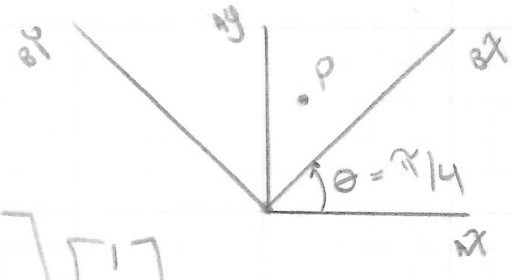
- Origin of coordinates are co-located
- Rotate one axis about z



${}_AP = {}_{AB}R\,{}_BP$

$${}_{AB}\hat{i} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \qquad {}_{AB}\hat{j} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

$$\Rightarrow \quad {}_{AB}R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$



A) Example

${}_AP = (1, 2)$

$${}_BP = {}_{BA}R\,{}_AP = \begin{bmatrix} \cos\pi/4 & -\sin\pi/4 \\ \sin\pi/4 & \cos\pi/4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

If we wanted to go back to coordinate frame A

$${}_AP = {}_{AB}R\,{}_BP = \begin{bmatrix} \cos\pi/4 & -\sin\pi/4 \\ \sin\pi/4 & \cos\pi/4 \end{bmatrix}$$

$$\Rightarrow {}_AP = {}_{AB}R\,({}_{BA}R\,{}_AP) \Rightarrow \underline{{}_{AB}R\,{}_{BA}R = I}$$

B) 2D Rotations : 2 Rotations

$_1P \rightarrow {}_3P$

$_3P = {}_3{}_1R(_{13}\theta)\,{}_1P$

$\Rightarrow {}_3{}_1R = \begin{bmatrix} \cos(_{12}\theta + {}_{23}\theta) & -\sin(_{12}\theta + {}_{23}\theta) \\ \sin(_{12}\theta + {}_{23}\theta) & \cos(_{12}\theta + {}_{23}\theta) \end{bmatrix}$ only works in 2D

or we could say

$_1P \rightarrow {}_2P \rightarrow {}_3P$

$\left.\begin{array}{l} {}_2P = {}_2{}_1R(_{12}\theta)\,{}_1P \\ {}_3P = {}_3{}_2R(_{23}\theta)\,{}_2P \end{array}\right\}$ Substitute in

$\Rightarrow {}_3P = {}_3{}_2R(_{23}\theta)\left[{}_2{}_1R(_{12}\theta)\,{}_1P\right] = \underbrace{{}_3{}_2R\,{}_2{}_1R}\,{}_1P$

A good way to make sure these are right is that the inside terms should match

C) Facts

- 2D rotation matrices are commutative
  - 3D is not

$A \cdot B = B \cdot A$

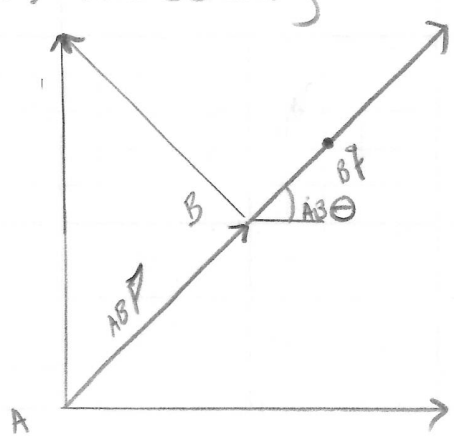
- We defined a fixed point rotation
  - ↳ Point is the same, different coords
- Rotation matrix is a part of special orthogonal group

III 2D Rotation and Translation

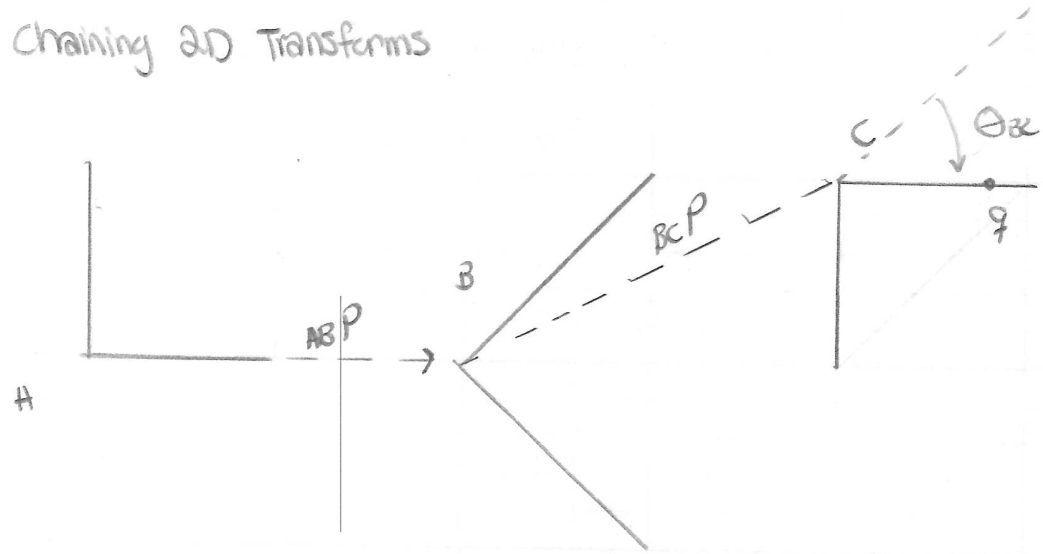
- This group is a part of special Euclidean group

$SE(2) = \{(p, R) : p \in \mathbb{R}^2,\ R \in SO(2)\}$

$^Aq = \underbrace{{}_A{}_BR\,{}_Bq}_{\text{Rotate}} + \underbrace{{}_AP}_{\text{Translate}}$



A) Chaining 2D Transforms



$$^{B}q = {}^{BC}R\,{}^{C}q + {}^{BC}P$$

$$^{A}q = {}^{AB}R\,{}^{B}q + {}^{AB}P$$

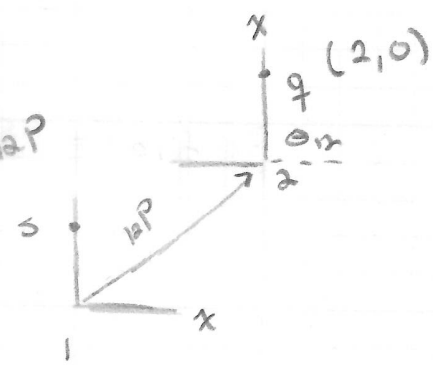
$$\Rightarrow {}^{A}q = {}^{AB}R\left({}^{BC}R\,{}^{C}q + {}^{BC}P\right) + {}^{AB}P$$

B) Taking Inverse

$${}^{1}q = {}^{12}P + {}^{12}R \cdot {}^{2}q \qquad {}^{2}q = {}^{21}P + {}^{21}R \cdot {}^{1}q$$

^{21}R in terms of $^{12}R \Rightarrow$ $\underline{^{21}R = {}^{12}R^{T}}$

^{21}P in terms of $^{12}P \Rightarrow {}^{21}P = -{}^{12}R^{T}\,{}^{12}P$



1) Example

$${}^{12}P = \begin{bmatrix} 9 \\ 2 \end{bmatrix} \qquad {}^{12}\theta = \pi/2$$

$${}^{12}R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$${}^{1}q = {}^{12}R\,{}^{2}q + {}^{12}P = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 9 \\ 2 \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

Find inverse parameter

$_{21}P = {}_{21}R_{12}P \qquad {}_{21}\theta = -\pi/2 \qquad {}_{21}R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 9 \\ 2 \end{bmatrix}$

$= \begin{bmatrix} -2 \\ 9 \end{bmatrix}$

IV 3D Rotation

$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} \qquad R_y(\beta) = \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix}$

$R_z(\alpha) = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

A) Euler Coordinates

There are 12 different representations (how you multiply them together)

$R(\gamma, \beta, \alpha) = R_z \cdot R_y \cdot R_x \Rightarrow$ 3D rotation matrices are not commutative

3D inverse is same as 2D

$_{12}R^{-1} = {}_{21}R = {}_{12}R^T$